

Paul Edelman

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EDUCATION

University of Delaware

Bachelor of Science in Computer Science, Bachelor of Arts in Applied Mathematics

Newark, DE

Aug. 2023 – May. 2027

EXPERIENCE

Software/Data Engineering Intern

May 2025 – Sep. 2025

AirGreen

New Castle, DE

- Developed and optimized Python-based data pipelines for ingesting, cleaning, and storing real-time IoT sensor data, ensuring high data integrity and throughput.
- Applied thermodynamic equations and statistical models to monitor and improve system efficiency, translating raw telemetry into actionable performance insights.
- Implemented SQL schema optimizations and indexing, reducing query latency by 40% in high-volume analytics workloads.
- Integrated predictive analytics into operational workflows, improving system uptime forecasts and enabling proactive maintenance.

Undergraduate Teaching Assistant

Aug. 2024 – May 2025

University of Delaware

Newark, DE

- Led weekly lab sessions and discussion groups focused on Python programming, data structures, and algorithmic problem-solving.
- Utilized tools such as Jupyter Notebook and GitHub to demonstrate coding best practices and collaborative version control.
- Developed supplementary course materials and debugging workshops, strengthening students' applied programming skills.
- Collaborated with faculty to refine curriculum and integrate industry-relevant software engineering practices.

Physics Lab Intern

Jun. 2022 – Aug. 2022

Delaware State University OSCAR Lab

Dover, DE

- Engineered multi-sensor acquisition using Raspberry Pi, Arduino, and Python to collect real-time environmental data.
- Processed and analyzed datasets with MATLAB, implementing signal filtering and statistical analysis pipelines.

PROJECTS

Real-time Collaborative Whiteboard | *React, WebSockets, Node.js, MongoDB*

Oct. 2025 – Jan. 2026

- Engineered a high-concurrency collaborative application using WebSockets to enable multi-user canvas synchronization with sub-100ms latency.
- Architected a scalable Node.js/Express backend and MongoDB schema to persist complex vector data and maintain state consistency across distributed client sessions.
- Optimized frontend performance by implementing Optimistic UI updates and custom state-management logic to handle real-time conflict resolution during simultaneous user inputs.
- Leveraged Vite and modular React components to reduce build times and ensure a responsive, low-overhead user experience during high event-stream volume.

Block-Smash | *JavaScript, Python, Flask, Node.js, PyTorch*

Jan. 2024 – Jun. 2024

- Leveraged JavaScript for a responsive front-end and Node.js to manage robust server-side functionalities.
- Built a scalable Python/Flask backend with real-time state management and low-latency client-server communication.
- Developed Smash-Bot, an AI game agent powered by PyTorch neural networks, to analyze gameplay patterns and execute high-scoring strategies.
- Gained experience in full-stack development, neural network design, and real-time data processing.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, SQL (Postgres), R, HTML/CSS

Backend & Data: Flask, FastAPI, Node.js, PostgreSQL optimization, Docker, CI/CD, Google Cloud Platform

Libraries: React, Socket.io, PyTorch, TensorFlow, pandas, NumPy, Matplotlib

Developer Tools: Git, VS Code, PyCharm, IntelliJ, Visual Studio